

Abstract:

The study of grand challenges such as climate change and sustainable use of natural capital demand increased collaboration across the sciences. The first steps for effective collaboration often focus on improving interoperability between observation and analyses methodologies. This later matures into standards. With the rapid evolution in technology, the time to create standards may lag well behind the technology in use. The ocean observation community - with regionally diverse members working in physics, chemistry, biology and engineering - is looking toward a more dynamic consensus-building approach to match the rapid pace of technological evolution.

This approach centers on the creation and use of “best practices” as guides for observations and analyses. Clearly communicating the methodologies used to generate data, in addition to having access to raw data itself, is essential for furthering reproducibility of results and facilitating multi-disciplinary analyses. To improve this communication, the ocean observation community has implemented an Ocean Best Practices System (OBPS). This System archives best practices on observations, data management, QA/QC, data analyses and applications. The System consists of a long-term OBPS repository at IODE enhanced with state-of-the-art semantic discovery and metadata indexing for improved access to best practices and, eventually, to the data associated with them. We have applied natural language processing and ontologies to link best practices to a common, machine readable knowledge model for improved interoperability. We are enhancing this prototype to give users more intuitive tools to discover best practices related to their environment, chemical compound, or sensor of interest. Further, we have created an academic forum, in “Frontiers in Marine Science”, for discussion of best practices and their applications. For broader outreach, we are also implementing a training program through the IODE Ocean Teacher project. Support from GOOS, IODE, JCOMM, national research institutes/infrastructures and others has made the system practical and vibrant. This presentation will discuss our technological basis and also the role of community in furthering the application of best practices. The extension of this work to other communities will also be examined.

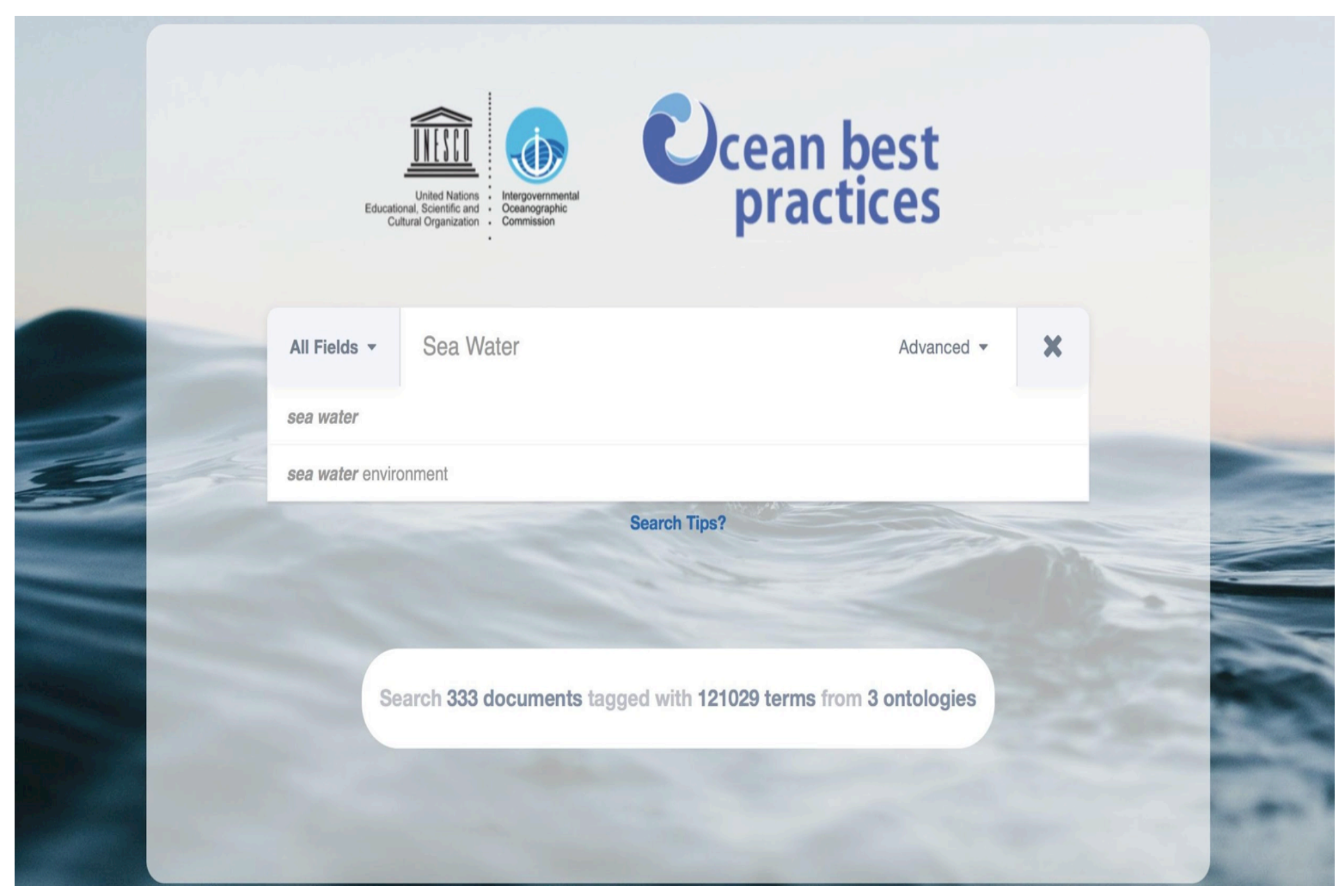
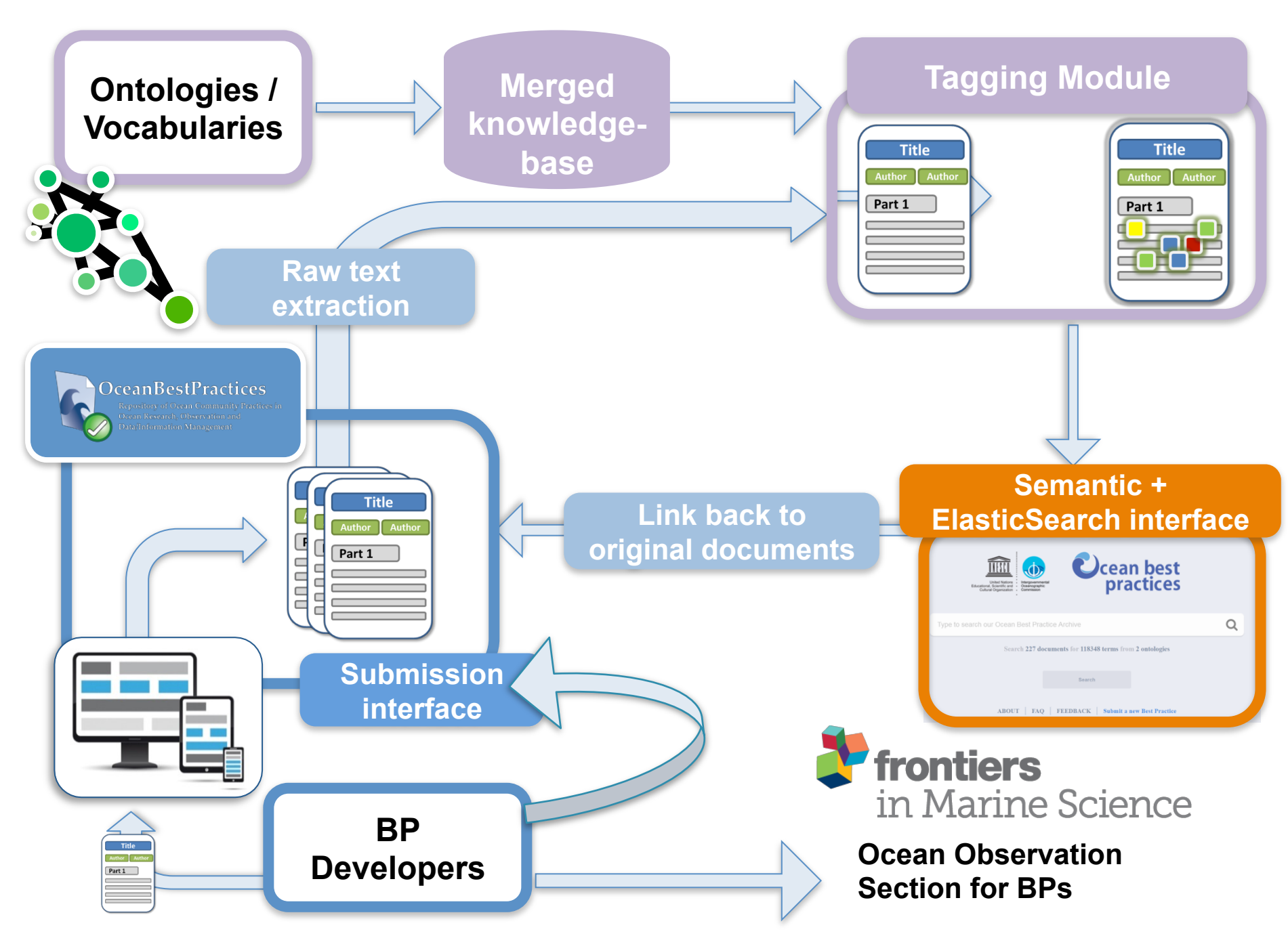
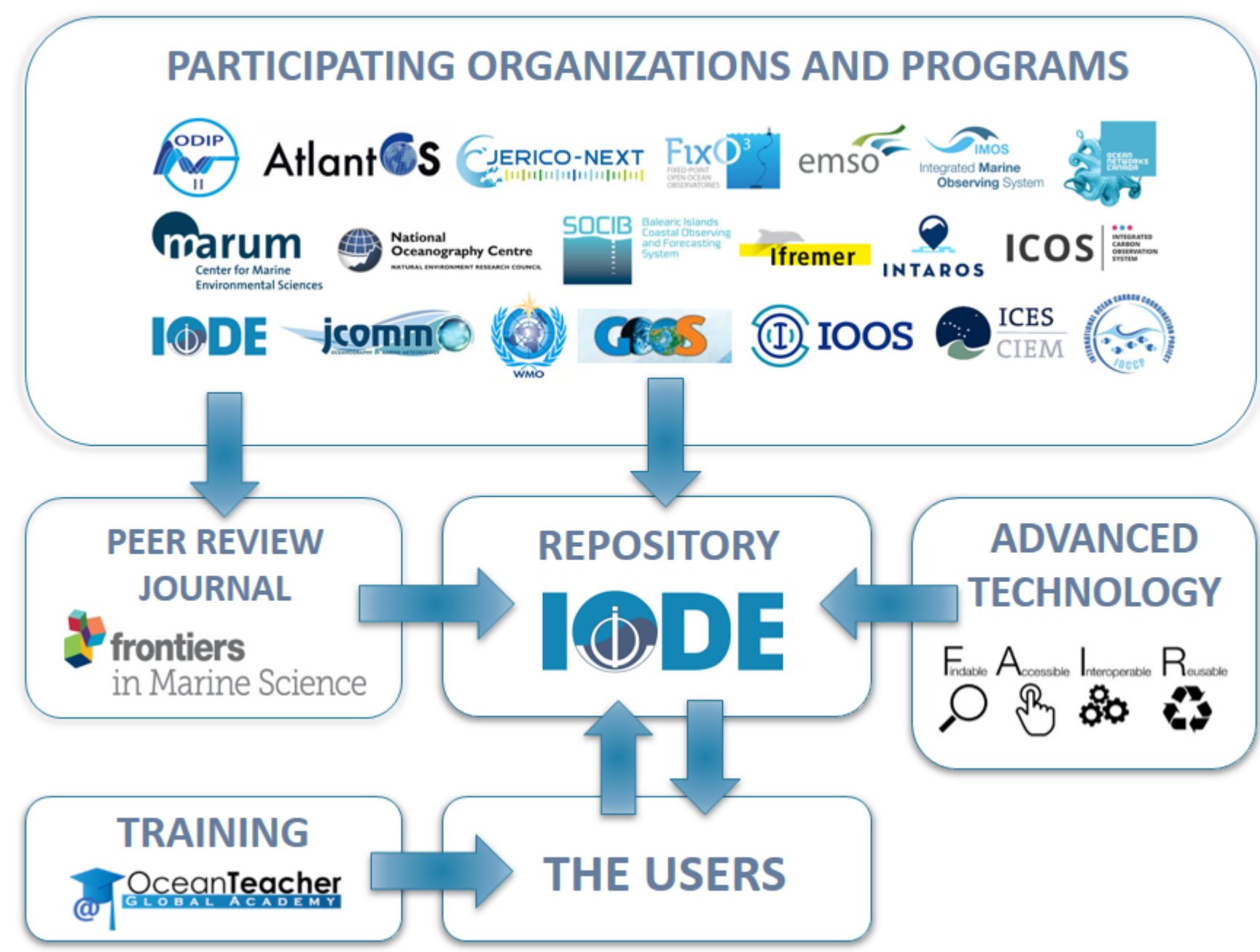
Oean Best Practices System

Vision: increase efficiency, reproducibility and interoperability of the entire ocean observing value chain by providing the ocean observing community with a unified, sustained and readily accessible knowledge base of interdisciplinary ocean best practices

Mission: provide coordinated and sustained global access to best practices in ocean observing to foster innovation and excellence, engaging the ocean communities in a joint and coordinated effort of producing, reviewing and sustaining best practices.

Best Practice Definition: A community best practice is a methodology that has repeatedly produced superior results relative to other methodologies with the same objective.

To be fully elevated to a best practice, a promising method needs to be adopted and employed by multiple organizations.





YOUR PARTICIPATION

The OBP repository and Frontiers Research Topic are now welcoming submissions and we encourage the ocean observing community to participate towards building a unified, sustained and readily accessible knowledge base of interdisciplinary ocean best practices.